Project Title

YouTube Analytics Data Cleaning using SQL

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2. Project Objective

The objective of this project is to help students understand how to clean and preprocess real-world YouTube analytics data using SQL.

Students will work on a raw YouTube dataset that contains multiple data quality issues such as:

- Missing values
- Duplicate records
- Invalid entries
- Inconsistent formats

By completing this project, students will learn how to transform messy video analytics data into a structured and usable format for performance analysis and reporting.

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3. Dataset Description

The dataset contains YouTube video analytics records with the following columns:

- **video_id** → Unique ID for each video
- **channel_name** → Name of the YouTube channel
- **title** → Video title
- **category** → Video category (Education / Gaming / Food / Travel / etc.)
- **views** → Number of views
- **likes** → Number of likes
- **dislikes** → Number of dislikes
- **comments** → Number of comments

- **publish_date** → Date when video was published
- **city** → Location of the channel/user
- **status** → Video status (Active / Inactive / Pending)

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4. Data Issues Present in Dataset

The dataset is intentionally dirty and contains the following problems:

🔹 **Structural Issues**

- Duplicate records (**video_id 9001 & 9011**)
- NULL / empty values in multiple columns
- Completely invalid rows (**empty, TestUser, Dummy, Unknown, NULL rows**)

🔹 **Data Type Issues**

- Views contain text values (**abc**)
- Dislikes contain text (**abc**)
- Likes, comments contain NULL values
- Mixed data types across columns

🔹 **Invalid Values**

- Negative values (**views = -20000, -1000**)
- Invalid numeric fields (**dislikes = abc**)
- Unrealistic entries (**TestUser, Dummy, Unknown**)
- Invalid placeholders (**???, NA**)

🔹 **Format Issues**

Category inconsistency:

- Education, education
- Gaming, gaming
- Food, food

City inconsistency:

- Hyderabad, hyd
- Bangalore, BLR
- Mumbai, mumbai

Status inconsistency:

- Active, active
- Pending
- Inactive

🔹 **Date Issues**

- Multiple formats:
    - 01-01-2024
    - 03-Jan-24
    - 07-Feb-24
- Invalid dates:
    - 99-99-2024
    - 00-00-0000
    - future_date
- Missing dates (NULL values)

🔹 **Special Character Issues**

- Channel names with symbols (**Tech@Guru, Cook!**)
- Titles with invalid values (**abc987**)

- Empty or NULL fields

🔹 **Analytics Data Issues**

- Non-numeric values in views/likes/comments
- Negative engagement values
- Missing engagement metrics
- Zero values that may indicate invalid data

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5. Tasks to Perform (For Students)

🔹 Task 1: Data Exploration

- Display all records
- Identify NULL values
- Find duplicate records
- Check incorrect data types
- Identify invalid entries

🔹 Task 2: Data Cleaning

Perform the following operations:

Handle Missing Values

- Replace NULL values where possible
- Remove rows with critical missing data

Remove Duplicates

- Identify duplicate videos using **video_id**

- Keep only one valid record

Fix Data Types

Convert columns into proper formats:

- **views → BIGINT**
- **likes → INT**
- **dislikes → INT**
- **comments → INT**
- **publish_date → DATE**

Standardize Data

- **category → Proper case (Education, Gaming, Food, Travel, etc.)**
- **status → (Active / Inactive / Pending)**
- **city → Proper case (Hyderabad, Bangalore, Mumbai, Delhi, Chennai)**
- **channel_name → Clean format**

Handle Invalid Values

- Remove negative values (views, likes, dislikes)
- Fix invalid numeric fields
- Remove rows with dummy values (**TestUser, Unknown, ???, NA**)

Clean Dates

- Convert all dates into **YYYY-MM-DD format**
- Fix invalid dates
- Handle NULL publish_date

Remove Special Characters

- Clean **channel_name** and **title**
- Remove symbols, numbers (where not meaningful), and extra spaces

🔹 Task 3: Create Clean Table

Create a new table:

👉 **clean_youtube_analytics**

Insert only cleaned and valid records into this table.

🔹 Task 4: Data Validation

Students must validate their cleaning:

- Count records before cleaning
- Count records after cleaning

Ensure:

- No duplicate **video_id**
- No NULL values in important columns
- Proper formats maintained

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6. Expected Outcome

After completing this project, students should be able to:

- Clean messy real-world YouTube datasets

- Handle missing and invalid data
- Write efficient SQL queries
- Standardize inconsistent data
- Prepare clean datasets for analytics and reporting

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7. Tools Required

- MySQL / PostgreSQL / SQL Server
- Excel (optional for viewing dataset)

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8. Important Instructions

- Do NOT modify the original table
- Always create a new cleaned table
- Do NOT delete data blindly — apply logic
- Maintain data consistency
- Use proper SQL functions

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9. Submission Requirements

Students must submit:

- SQL queries used for cleaning
- Cleaned dataset (table output)

Before vs After Comparison:

- Record count
- Number of duplicates removed
- Number of NULLs handled
- Invalid values corrected